DRAWINGS

AI-Based Agricultural Marketplace for Farmer–Buyer Matching and Net Realization Optimization

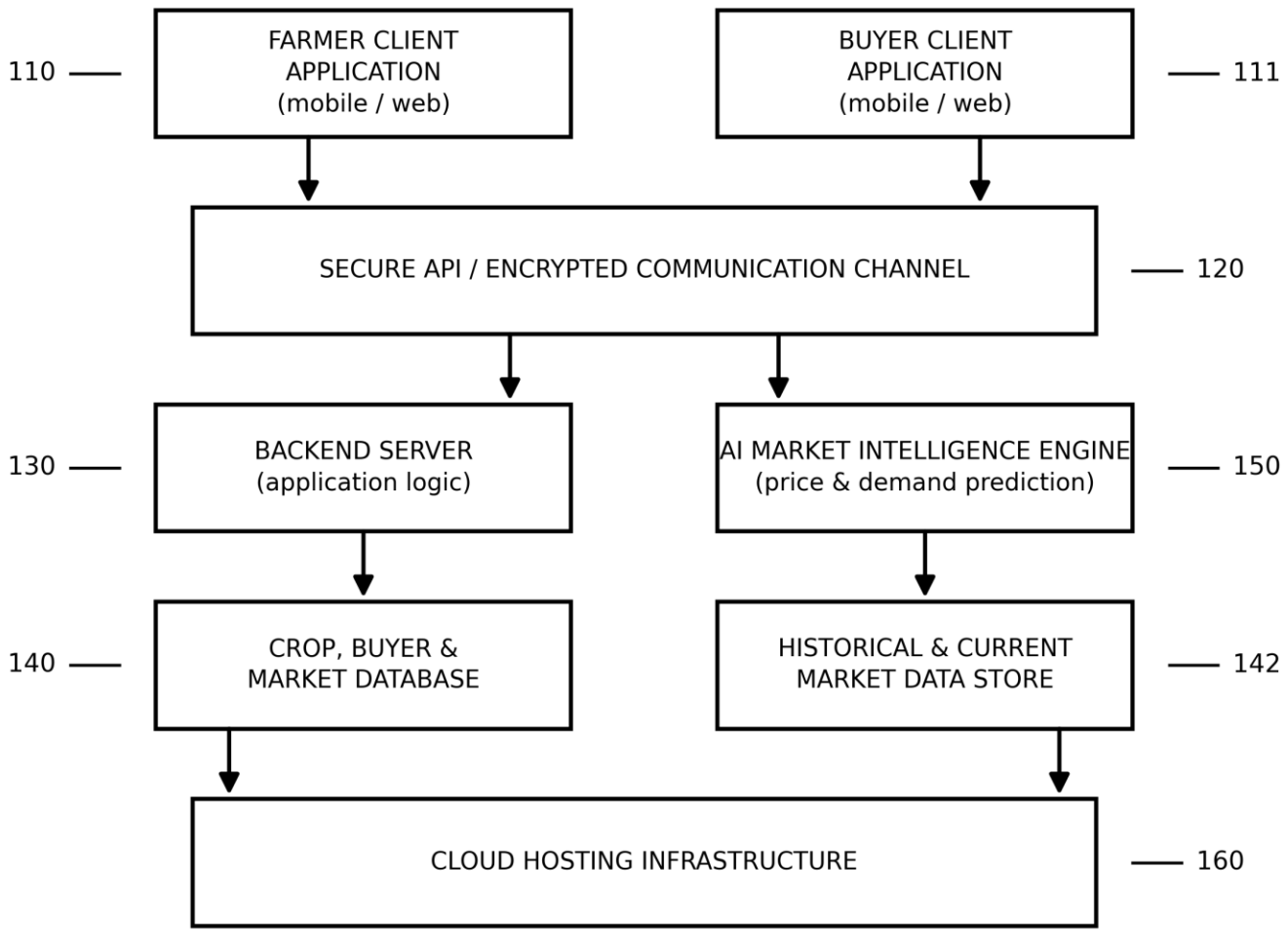

110
FARMER CLIENT APPLICATION (mobile / web)
BUYER CLIENT APPLICATION (mobile / web)
111
SECURE API / ENCRYPTED COMMUNICATION CHANNEL
120
130
BACKEND SERVER (application logic)
AI MARKET INTELLIGENCE ENGINE (price & demand prediction)
150
140
CROP, BUYER & MARKET DATABASE
HISTORICAL & CURRENT MARKET DATA STORE
142
CLOUD HOSTING INFRASTRUCTURE
160

FIG. 1

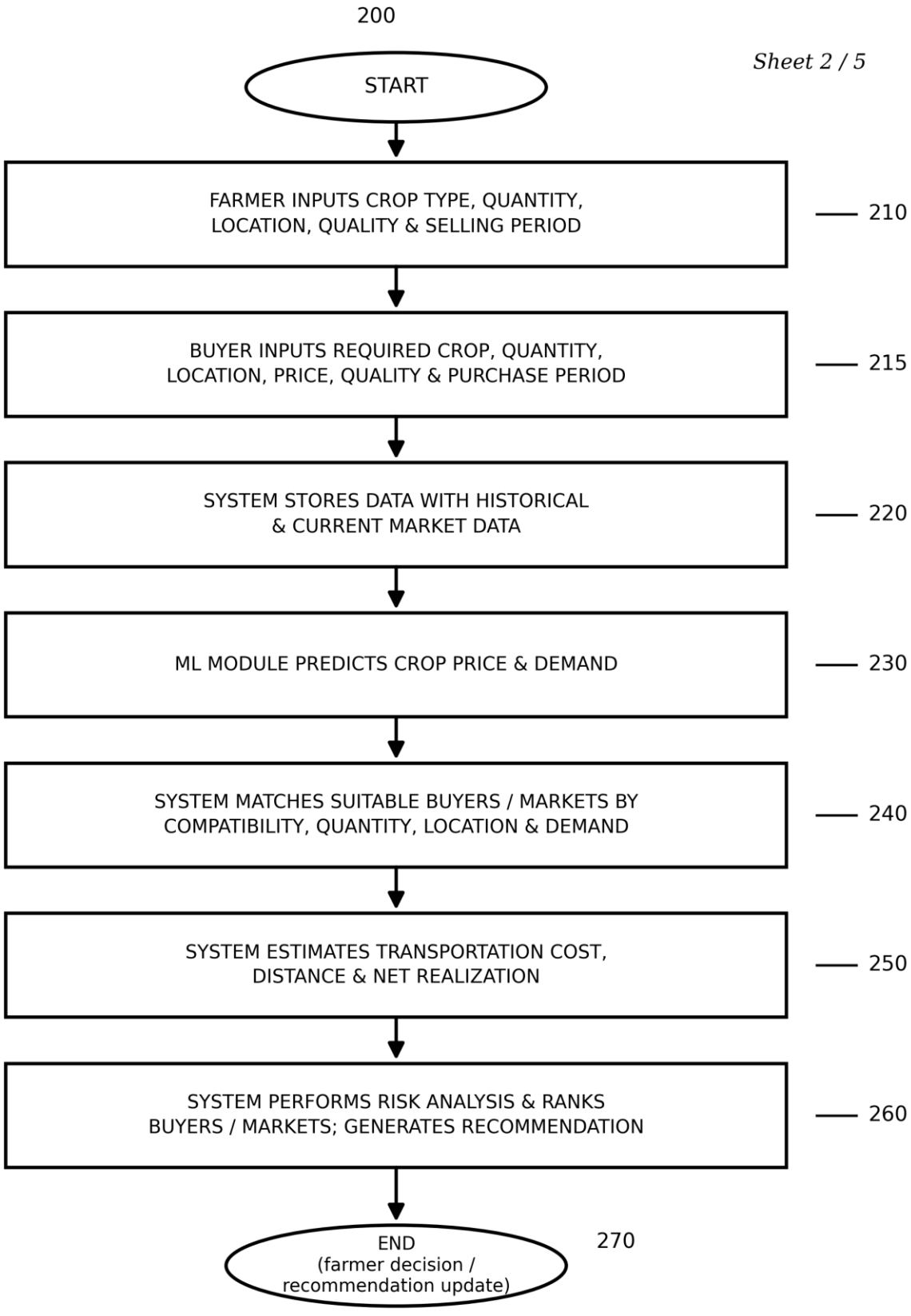

200
START
FARMER INPUTS CROP TYPE, QUANTITY, LOCATION, QUALITY & SELLING PERIOD
210
BUYER INPUTS REQUIRED CROP, QUANTITY, LOCATION, PRICE, QUALITY & PURCHASE PERIOD
215
SYSTEM STORES DATA WITH HISTORICAL & CURRENT MARKET DATA
220
ML MODULE PREDICTS CROP PRICE & DEMAND
230
SYSTEM MATCHES SUITABLE BUYERS / MARKETS BY COMPATIBILITY, QUANTITY, LOCATION & DEMAND
240
SYSTEM ESTIMATES TRANSPORTATION COST, DISTANCE & NET REALIZATION
250
SYSTEM PERFORMS RISK ANALYSIS & RANKS BUYERS / MARKETS; GENERATES RECOMMENDATION
260
END (farmer decision / recommendation update)
270

FIG. 2

CROP SELLING OPPORTUNITY RECORD

310

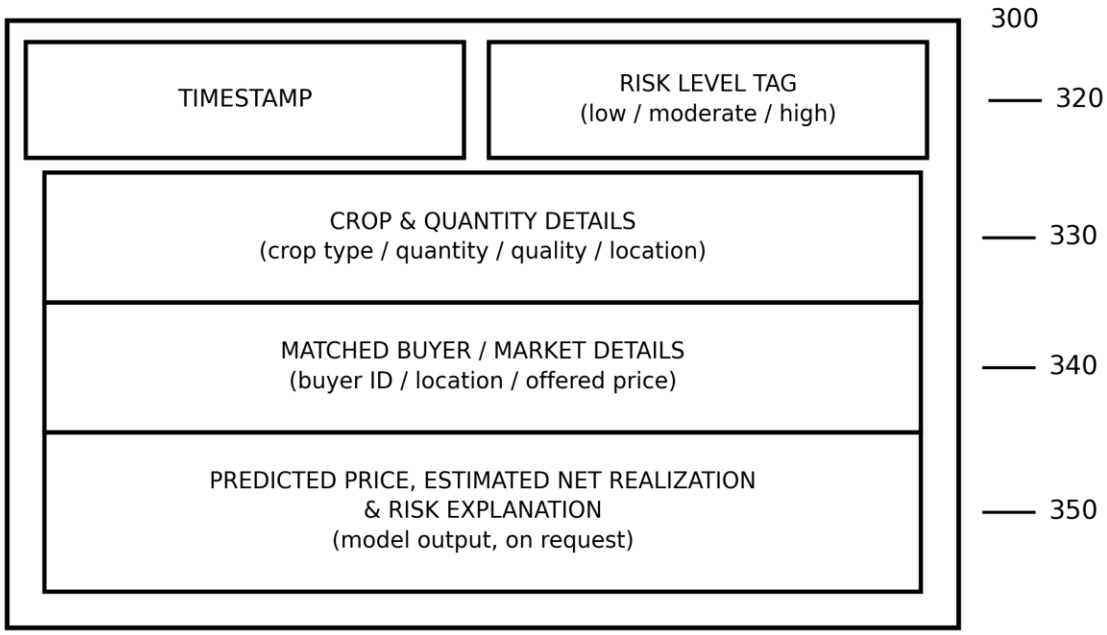


FIG. 3

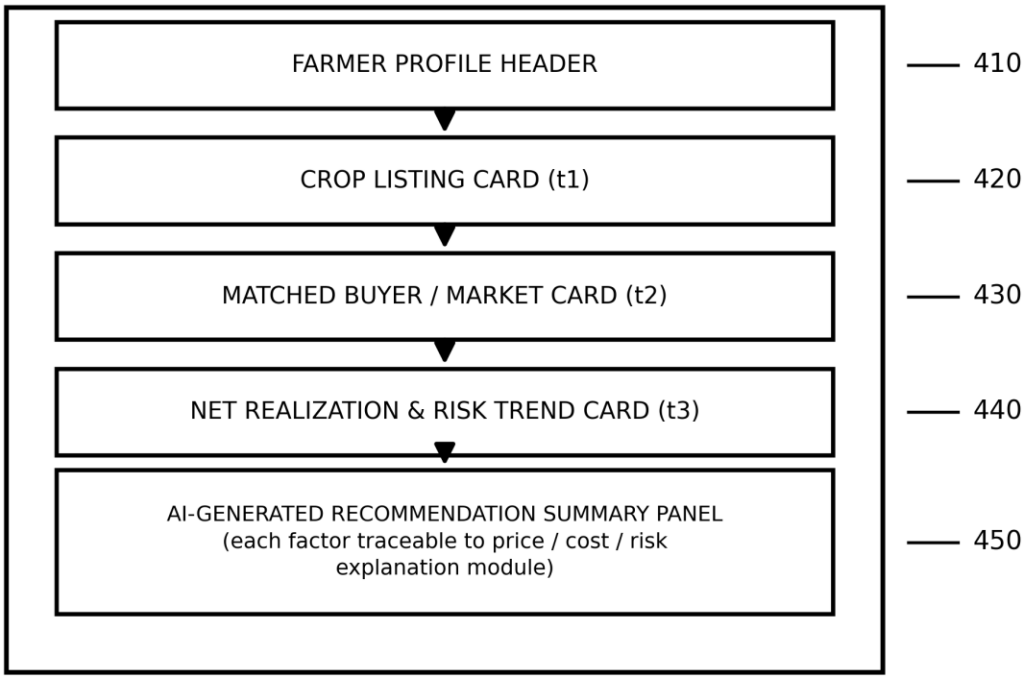

FARMER PROFILE HEADER
410
CROP LISTING CARD (t1)
420
MATCHED BUYER / MARKET CARD (t2)
430
NET REALIZATION & RISK TREND CARD (t3)
440
AI-GENERATED RECOMMENDATION SUMMARY PANEL
(each factor traceable to price / cost / risk
explanation module)
450

FIG. 4

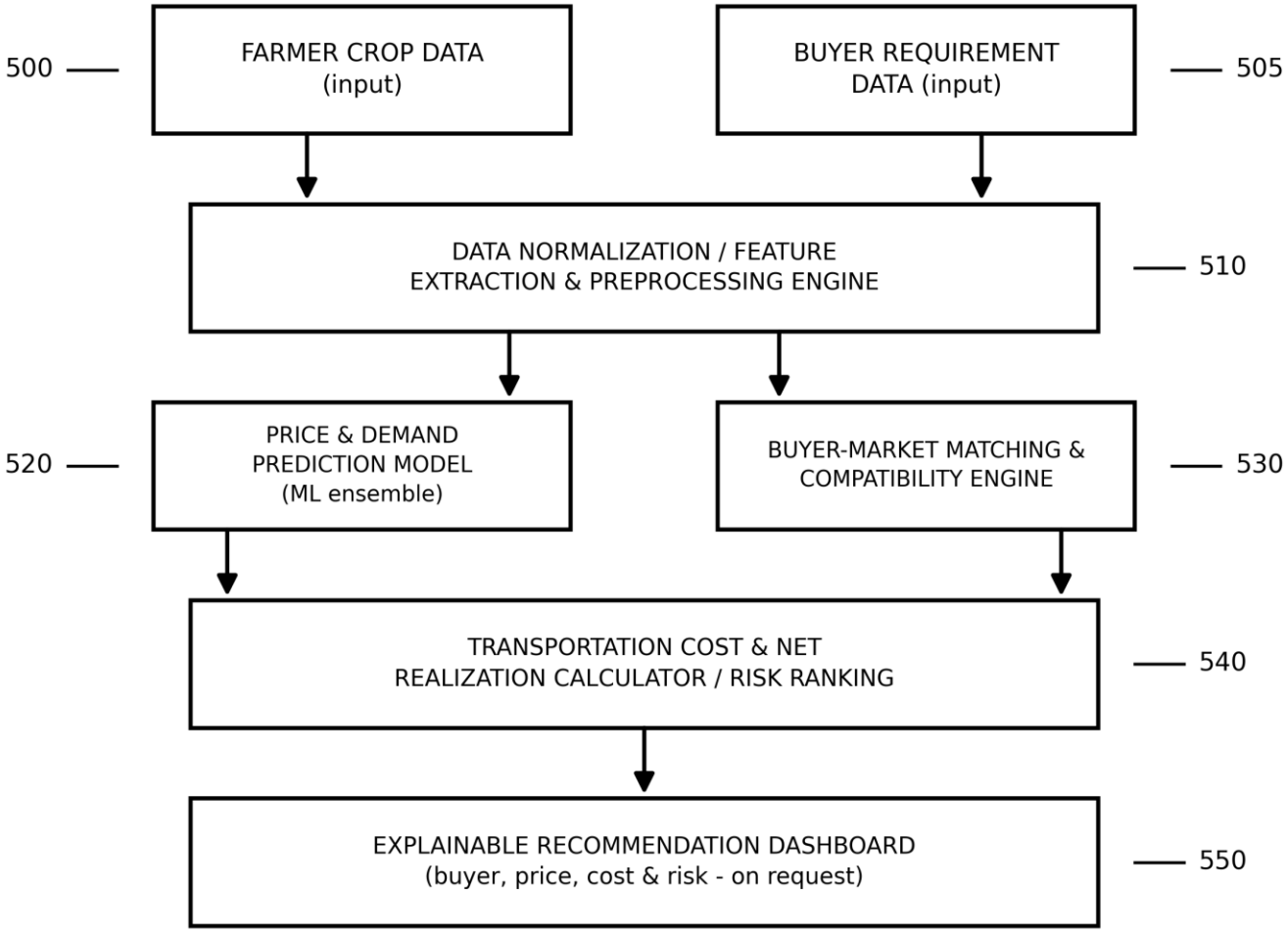

500
FARMER CROP DATA
(input)
BUYER REQUIREMENT
DATA (input)
505
DATA NORMALIZATION / FEATURE
EXTRACTION & PREPROCESSING ENGINE
510
520
PRICE & DEMAND
PREDICTION MODEL
(ML ensemble)
BUYER-MARKET MATCHING &
COMPATIBILITY ENGINE
530
TRANSPORTATION COST & NET
REALIZATION CALCULATOR / RISK RANKING
540
EXPLAINABLE RECOMMENDATION DASHBOARD
(buyer, price, cost & risk - on request)
550

FIG. 5